

List of the Projection in Geo-Spatial Data

S.No	Projection Name	Properties	What They Preserve	Formula	Founded In	Founded By
1	Eckert I	Pseudocylindrical projection with curved parallels and equally spaced meridians	Neither area nor shape	$X = \lambda(1 + \cos(\phi))/2, Y = \phi$	1906	Max Eckert-Greifendorff
2	Eckert II	Pseudocylindrical projection with straight parallels	Neither area nor shape	$X = \frac{\lambda\sqrt{4 - 3\sin^2(\phi)}}{2}, Y = \phi$	1906	Max Eckert-Greifendorff
3	Eckert III	Pseudocylindrical projection with curved meridians, attempting to balance distortion	Neither area nor shape	$X = \frac{2}{\sqrt{\pi}}\lambda\cos(\phi), Y = \phi$	1906	Max Eckert-Greifendorff
4	Equidistant Conic	Conic projection, meridians are straight lines, parallels are arcs	Distance along meridians	$X = r\sin(\theta), Y = r_0 - r\cos(\theta)$	Ancient	Hipparchus, Ptolemy
5	Equidistant Cylindrical	Cylindrical projection, equally spaced meridians and parallels	Distance along meridians	$X = \lambda, Y = \phi$	Ancient	Unknown, commonly used by Greeks
6	Equidistant Cylindrical ARC System	Cylindrical, used for mapping global datasets in geospatial applications	Equidistant properties	Similar to Equidistant Cylindrical projection formula	Modern	Modern GIS Systems

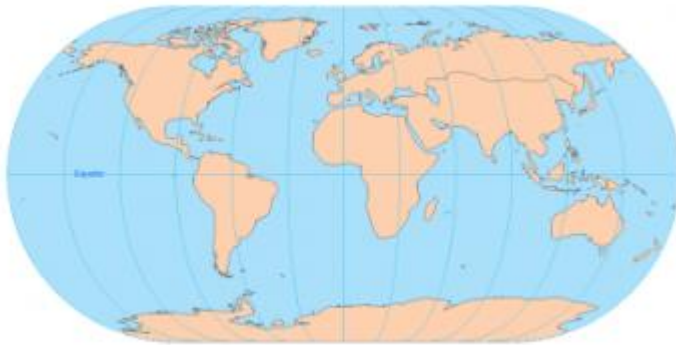
7	Flat Polar Quartic	Pseudocylindrical, four-lobed projection, minimizes distortion at the poles	Equal-area	Uses quartic curves for latitude lines, more complex derivation	1952	Karl Siemon
8	Fuller (Dymaxion Map)	Polyhedral projection based on an icosahedron, minimizes distortion by breaking map into triangles	Neither area nor shape	Irregular projection, no simple formula, geometric approximation	1943	R. Buckminster Fuller
9	Gall Stereographic	Cylindrical projection, compromise between Mercator and other projections	Neither area nor shape	$X = \lambda \cos(45^\circ)$, $Y = 2\phi \cos(45^\circ)$	1855	James Gall
10	Goode Homolosine	Hybrid projection, combines Sinusoidal and Mollweide projections	Area (Equal-area)	Composite of Sinusoidal and Mollweide formulas	1923	John Paul Goode

Visualization:

1. Eckert-1



2. Eckert-2



3. Eckert-3



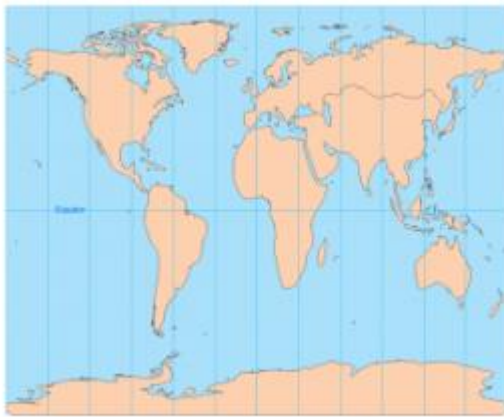
4. Equidistant Conic



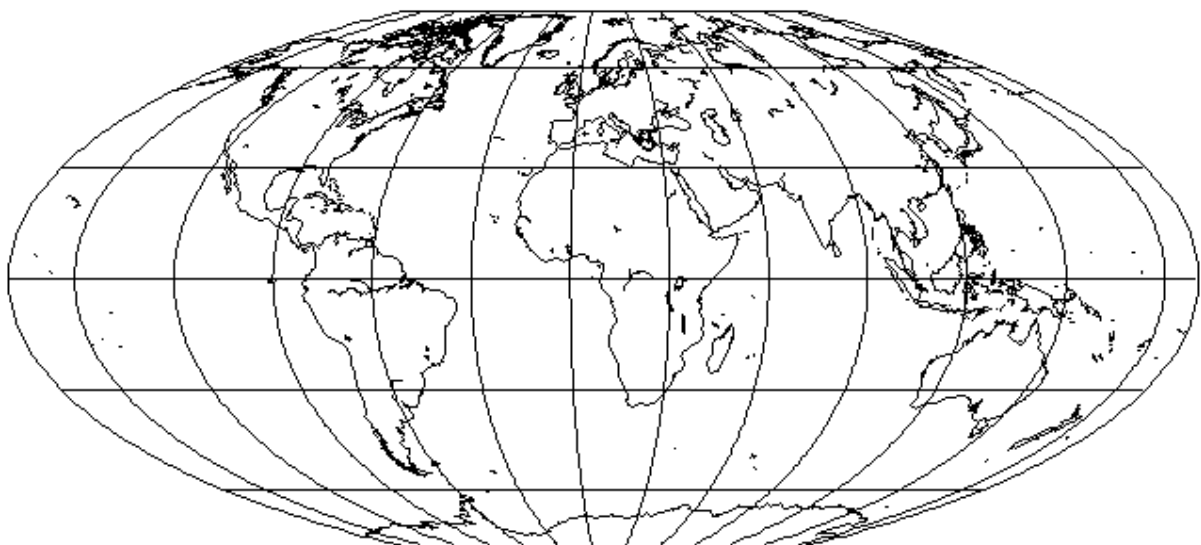
5. Equidistant Cylindrical



6. Equidistant Cylindrical ARC System



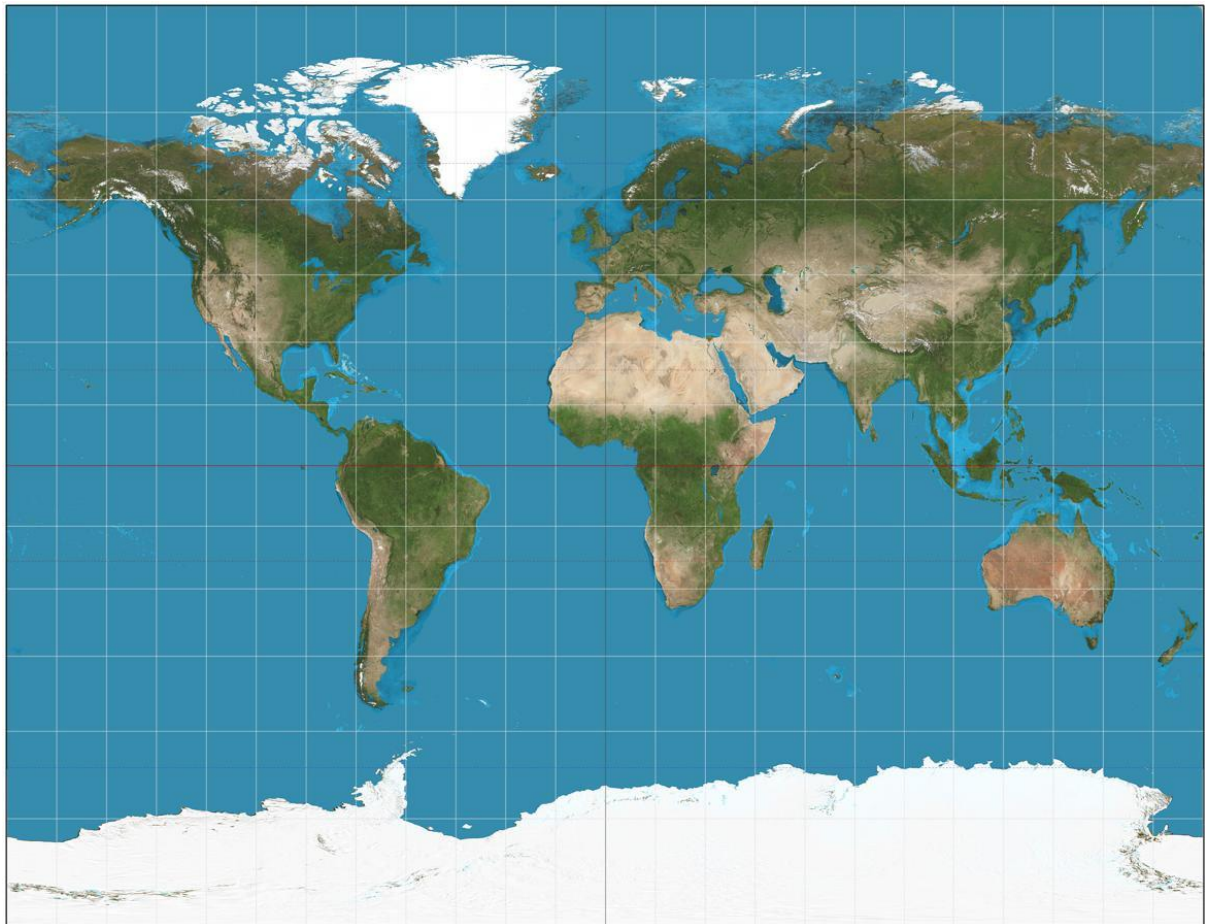
7. Flat Polar Quartic



8. Fuller



9. Gall Stereographic



10.Goode Homolosine Land

